

## Static Gesture Classification — Run Report

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## Data paths

|                       |                                                                        |
|-----------------------|------------------------------------------------------------------------|
| Training data path 1  | /home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Static · 30 trials       |
| Training data path 2  | /home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Static · 30 trials       |
| Training data path 3  | /home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Static · 30 trials     |
| Training data path 4  | /home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Static · 30 trials     |
| Training data path 5  | /home/jestin/ThesisRepo/ML/NewTestData/5_Georgia/Static · 30 trials    |
| Training data path 6  | /home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Static · 30 trials      |
| Training data path 7  | /home/jestin/ThesisRepo/ML/NewTestData/7_Henry/Static · 30 trials      |
| Training data path 8  | /home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static · 31 trials     |
| Training data path 9  | /home/jestin/ThesisRepo/ML/NewTestData/10_Josh/Static · 30 trials      |
| Training data path 10 | /home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Static · 31 trials     |
| Training data path 11 | /home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Static · 30 trials    |
| Training data path 12 | /home/jestin/ThesisRepo/ML/NewTestData/13_Nat/Static · 32 trials       |
| Training data path 13 | /home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Static · 30 trials |
| Training data path 14 | /home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Static · 0 trials       |
| Training data path 15 | /home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Static · 0 trials      |
| Training data path 16 | /home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Static · 30 trials    |
| Total training paths  | 16 · 424 trials total                                                  |
| External test root    | /home/jestin/ThesisRepo/ML/NewTestData/9_Joselyn/Static                |
| Report output dir     | /home/jestin/ThesisRepo/ML/saved_reports                               |

## Sensor selection

|                       |                                                                                                     |
|-----------------------|-----------------------------------------------------------------------------------------------------|
| <b>Hands</b>          | left, right                                                                                         |
| <b>Segments</b>       | palm_mid, thumb, index, middle, ring, pinky, wrist                                                  |
| <b>Modalities</b>     | YPR, Accel, Flex                                                                                    |
| <b>Sensor columns</b> | 176 channels                                                                                        |
| <b>Classes</b>        | 6: Double_Closed_Fist, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman |

## Preprocessing, features & split

|                             |                                             |
|-----------------------------|---------------------------------------------|
| Resample steps              | 90                                          |
| Butterworth                 | off · cutoff 10.0 Hz · order 4 · fs 30.0 Hz |
| Normalisation               | minmax                                      |
| Feature mode                | stats+fft → feature dim 9504                |
| Test size · seed · CV folds | 0.2 · random_state=42 · 10-fold stratified  |
| Sample counts               | train (post-aug)=339 · test=85              |

### Augmentation (training only)

|                        |                        |
|------------------------|------------------------|
| Enabled                | no                     |
| Copies per train trial | 1                      |
| Jitter                 | off · sigma=0.01       |
| Amplitude scale        | on · range=(0.8, 1.2)  |
| Time warp              | off · range=(0.9, 1.1) |

## Classifiers — cross-validation & hold-out test

| Algorithm                       | CV mean (10-fold) | CV std | Hold-out test acc |
|---------------------------------|-------------------|--------|-------------------|
| Stacking (SVM+RF+KNN+LogReg)    | 0.9851            | 0.0201 | 1.0000            |
| SVM (Linear)                    | 0.9615            | 0.0377 | 0.9882            |
| Random Forest                   | 0.9912            | 0.0135 | 0.9882            |
| Voting Soft (SVM+RF+KNN+LogReg) | 0.9675            | 0.0279 | 0.9882            |
| SVM (RBF)                       | 0.9497            | 0.0376 | 0.9765            |
| Logistic Regression             | 0.9556            | 0.0332 | 0.9765            |
| LDA                             | 0.8967            | 0.0498 | 0.9294            |
| KNN                             | 0.7816            | 0.0532 | 0.8353            |

### Per-class hold-out F1

[illegible]

